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Eliciting Attribute-Level User Needs From Online Reviews With Deep Language Models and Information Extraction

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Eliciting user needs for individual components and features of a product or a service on a large scale is a key requirement for innovative design. Synthesizing data as an initial discovery phase of a design process is usually accomplished with a small number of participants, employing qualitative research methods such as observations, focus groups, and interviews. This leaves an entire swath of pertinent user behavior, preferences, and opinions not captured. Sentiment analysis is a key enabler for large-scale need finding from online user reviews generated on a regular basis. A major limitation of current sentiment analysis approaches used in design sciences, however, is the need for laborious labeling and annotation of large review datasets for training, which in turn hinders their scalability and transferability across different domains. This article proposes an efficient and scalable methodology for automated and large-scale elicitation of attribute-level user needs. The methodology builds on the state-of-the-art pretrained deep language model, BERT (Bidirectional Encoder Representations from Transformers), with new convolutional net and named entity recognition (NER) layers for extracting attribute, description, and sentiment words from online user review corpora. The machine translation algorithm BLEU (BiLingual Evaluation Understudy) is utilized to extract need expressions in the form of predefined part-of-speech combinations (e.g., adjective–noun, verb–noun). Numerical experiments are conducted on a large dataset scraped from a major e-commerce retail store for apparel and footwear to demonstrate the performance, feasibility, and potentials of the developed methodology. [DOI: 10.1115/1.4048819]

Keywords: user-centered design, e-commerce, natural language processing, named entity recognition, BERT, data-driven design, machine learning, product design

1 Introduction

The rapid growth of e-commerce platforms provides an unprecedented source of information about the needs of large and diverse groups of users on a multitude of products and services. Recent surveys show that online reviews influence the purchase decisions for 93% of users [1], and 91% of users trust online reviews as much as personal recommendations [2]. Yet, there is a lack of efficient and accurate mechanisms to assist designers in identifying the needs of individual users on a large scale from online reviews. The elicitation and incorporation of user needs in the design process have been proven effective in the overall success of new product/service development processes [3] by increasing the quantity and the quality of ideas at the front-end of the design process [4,5]. In view of shrinking product development cycles and pursuit of better outcomes, there is a substantial opportunity to improve the front-end of design innovation processes by collecting and learning from new sources of "big data" available on a variety of e-commerce platforms. This article develops a novel artificial intelligence (AI) methodology built upon deep language representation models [6,7] and information extraction methods [8–10] for large-scale elicitation of user needs from online reviews in an efficient and scalable fashion. The methodology is tested and validated on large dataset scraped from a major apparel and footwear e-commerce store.

1.1 Research Gap and Motivation.

Understanding of user needs is a preliminary step in early-stage product development [11]. The majority of existing need finding and elicitation approaches are based primarily on qualitative analysis of former designs [12,13], surveys and focus group studies [14,15], and web-based configurators [16,17], which are inherently biased due to targeting a small fraction of users population and product instances, and being limited to structured inquiries. The inability of users to directly articulate their needs [18] and the confirmation bias [19] of designers further exacerbate this gap. As a result, these limitations have so far hindered industrial adoption of mass customization and personalization paradigms due to the inherent extensive economic and operational limitations [20]. Furthermore, due to the limited ability to address the voices of large groups of customers, the gap between "what is designed" and "what is desired" lingers notwithstanding the knowledge and expertise of the design team.

Sentiment analysis has become a key enabler for *large-scale* need finding by allowing to extract the opinions of myriad users from e-commerce and social media platforms [21–25]. Sentiment analysis is the process of identifying the subjective opinion of an opinion holder (e.g., user) for a target (e.g., a product attribute) from an unstructured text (e.g., a product review) [24]. Over the past two decades, extensive research has been conducted on the sentiment analysis, and a multitude of rule-based and statistical natural language processing (NLP) methods have been developed for extracting the preferences, opinions, and needs of large groups of users [11,13,26–37]. Nevertheless, major drawbacks remain to be addressed: current methods are crucially dependent on large manually labeled, tagged, and annotated data (see, e.g., Refs. [38,39]). This, in turn, limits their practical use due to their laborious nature and the lack of scalability and transferability to other

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domains. This work takes a preliminary step toward addressing this challenge and identifying several fundamental questions to be addressed in future research.

1.2 Objectives and Outline. This article develops a domain-agnostic methodology for eliciting the attribute-level needs of individual users from the reviews they post on e-commerce platforms on a regular basis. An *attribute*, in this context, may refer to any component or feature of a product or a service. The methodology builds on pretrained deep language representation model BERT (Bidirectional Encoder Representations from Transformers) by Google AI Language [7] and on named entity recognition (NER) [8–10] for efficient and scalable transfer across different need finding domains [40–47]. Language representation models such as Google BERT and OpenAI Generative Pretrained Transformer (GPT) [6] are generic NLP models pretrained on massive textual data (e.g., BooksCorpus [48] and English Wikipedia). The purpose of these models is to reduce the need for domain-specific architectures, large training data, and the learning of parameters from scratch for each new task [6,7,49,50]. The fine-tuning of these language representation models, however, requires domain-specific labeled data yet on a much smaller scale compared to traditional NLP methods. The BERT-NER methodology comprises the following objectives (Fig. 1):

(1) *Product attribute specification.* Build an attribute lexicon by manually extracting product descriptions from online product catalogs and glossaries grouping them into a set of attributes (e.g., for sneakers: color, energy return, permeability, weight, stability, durability).
(2) *Named entity recognition.* Develop an NER model to locate named entities in a review and classify them into four categories of attribute, description, sentiment, and background word.
(3) *Deep language model.* Fine-tune and train a BERT model on a small labeled dataset for an *imbalanced classification* task based on the labels generated by the NER model (*Note*: The classification task is imbalanced because the majority of the named entities in a review are typically "background" words).
(4) *Attribute-level need elicitation.* Elicit attribute-level expressions of user needs by postprocessing the BERT-NER outputs through (i) applying a preexisting domain-specific attribute lexicon and a semantic similarity measure [51] to extract the most related and frequent attributes and (ii) part-of-speech (POS) tagging and identifying the attribute-level need expressions using the BLEU (BiLingual Evaluation Understudy) score [52].

The remainder of this article is organized as follows. Section 2 provides an extensive summary of related topics including attribute extraction, NER, and the evolution of language models. Section 3 discusses the details of the BERT-NER methodology. Section 4 presents the experimental results, analyses, and implications of the developed methodology. Section 5 provides concluding remarks and several directions for future research.

2 Background

This section presents a brief yet comprehensive review of the related previous work. Readers familiar with these topics may skip this section without losing the scope of this work.

2.1 Attribute Extraction Methods. The computational study of people's opinions, preferences, emotions, or attitudes toward the attributes of a product or service, event, or subject is known as sentiment analysis [30]. This relatively young area of NLP research has recently gained tremendous attention in the design science community due to its unprecedented ability to quantitatively measure user satisfaction for a product or a service on a large scale from online reviews [22,24,25,30,31,53,54]. The process typically involves identifying the opinion, the opinion holder, the attribute associated with the opinion, and the polarity orientation of the opinion to the attribute [55,56]. A critical step in the sentiment analysis process is the efficient and accurate extraction of attributes, which is the focus of the BERT-NER methodology developed in this article. This is a particularly challenging problem to solve for two reasons. First, users may choose different words or expressions to refer to the same standard attribute of a product or a service.

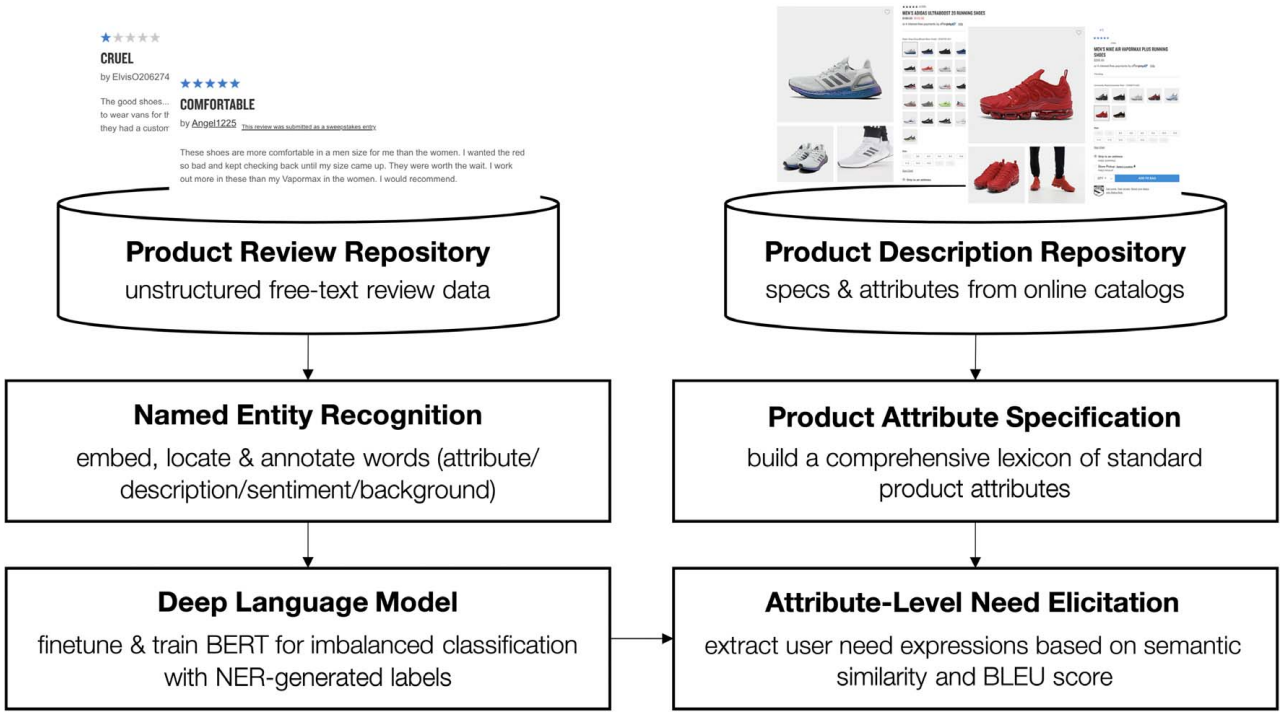


Fig. 1 Overview of the BERT-NER methodology for large-scale elicitation of attribute-level user needs from online reviews

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Second, users occasionally discuss implicit attributes that may not be standard, obvious, or easy to find [39], but may contain crucial and potentially transformative information for new product design and development processes.

Various methods have been proposed in the literature for attribute extraction from the unstructured review data [57]. The majority of these methods require manual labeling and annotation of data. Subjective determination is one of the early approaches for identifying product-attribute words [58], which requires laborious annotation of the review data that may also be prone to bias. Traditional supervised learning methods such as conditional random fields have also been used for this purpose [59,60], which by nature require large volumes of manually labeled training data. Rule-based methods are a different approach used in many attribute extraction studies [39,61,62], which require hand-crafted rules and are limited to the specific domain for which the rules are defined. Another category of attribute extraction methods involves assigning proper weights to each of the candidate attribute words, including term frequency-inverse document frequency weighting schemes [63] and topic modeling techniques such as latent Dirichlet allocation [64,65]. The former may lead to generating several unrelated nouns that require tedious postprocessing of the results, while the latter requires to define a predetermined number of topics to generate, which may cause undesirable expert bias in the results. POS patterns have also been used in some studies [66,67]; however, these methods also require subjective POS tagging of the training data.

2.2 Named Entity Recognition. NER is an information extraction method introduced in the 1990s [68] for detecting named entities and classifying references to those entities in predefined categories. Nadeau and Sekine [69] present an extensive review of NER methods and classify them into three categories of supervised, semisupervise, and unsupervised learning. The majority of NER methods are classified as supervised learning, which typically require large annotated corpora for learning named the entity detection and disambiguation rules. Common supervised learning methods for NER include hidden Markov models [70], decision trees [71], maximum entropy models [72], support vector machines [73], and conditional random fields [74]. Bootstrapping is the major semi-supervised learning method used for NER, which requires a smaller set of annotated seeds for initiating the learning process and then reapplying the learned rules to newly found examples [75]. Clustering is the main unsupervised learning method used for NER (e.g., Refs. [76,77]), where named entities are grouped into clusters based on their semantic similarities calculated using lexical resources such as WordNet [51].

2.3 Pretrained Language Models. Language models encode natural language into numbers by estimating the relative likelihood of words and phrases in a text. These models are a major building block of many NLP tasks. Language models were initially introduced in the early 20th century with feedforward neural networks [78] and have since then evolved into many successful methods and technologies. One of the most popular language models is GoogleNews-300 Word2Vec pretrained model [79]—a two-layer neural network that transforms text into vectors. The model is trained by a skip-gram model with hierarchical softmax output [80] and can transfer words into n dimensional vectors, where n is another training parameter (e.g., the size of the hidden layer). GloVe (Global Vectors for Word Representation) is another well-known unsupervised language model based on a co-occurrence matrix [81]. Language models have played a pivotal role in NLP research in recent years [82]. With the rapid growth of deep learning, researchers have begun to pretrain language models on large corpora. GPT by OpenAI [6] and BERT by Google AI Language [7] are two models that use "transformers" to train deep neural networks on massive datasets such as BooksCorpus [48] and English Wikipedia. Users merely require to "fine-tune" these capable deep language models for their particular NLP task (e.g., by adding a single layer atop), proven to achieve state-of-the-art performance in several NLP tasks. In this article, we utilize BERT as our base deep language model and fine-tune it for the attribute and sentiment extraction tasks with a minibatch size of 100, over 200 epochs that take around 30 min to run on CUDA.

3 The BERT-NER Methodology

This section discusses the BERT-NER methodology (see Fig. 1). Preliminary definitions and assumptions are presented first, followed by the structure and the formulation of the BERT model and the NER layer. The BLEU score [52] is then used as a postprocessor for eliciting attribute-level user needs from the BERT-NER results.

3.1 Preliminaries. *Assumptions.* The methodology is built upon the fundamental assumption that an attribute word is a noun [57], representing a part (e.g., sole) or a feature (e.g., comfort) of the product [54]. Since the review corpus contains thousands of distinct nouns, two additional assumptions have also been made: (i) a product attribute is a frequent domain-specific noun [83] and (ii) users apply a range of synonymous terms to refer to the same attribute, some of which may be uncommon/infrequent.

Definitions and Preliminary Steps. The definitions and preliminary steps for building the BERT and NER models are as follows:

(1) *Labeling set*. Four types of labels are used to differentiate between the words in the reviews: "NA" as a background word, "S" as a sentiment word, "A" as an attribute word, and "D" as the description of those attributes. For example, the words in the sentence "Great shoes, look fantastic and very comfortable, love the upper feeling" are labeled as {S, A, A, S, NA, NA, S, S, NA, A, D}, in that order. The labels are generated manually for a small dataset (130 labeled reviews total: 90 for training, 20 for validation, and 20 for testing).
(2) *Attribute lexicon*. A domain-specific attribute lexicon is used, which has been developed based on existing online product catalogs and attribute dictionaries. The attribute lexicon includes descriptions of sneakers manually extracted from around 1000 online product reviews. The manually generated attribute lexicon comprises a total of roughly 500 descriptions grouped into 23 main attributes (e.g., color, energy return, permeability, weight, stability, and durability).
(3) *Preprocessing*. Reviews sometimes contain meaningless or irrelevant information. The preprocessing step translates the content to restored language and eliminates the remaining content that cannot be restored.
(4) *Word embedding*. Word embedding is a language modeling method that transfers words into high-dimensional vectors. The embedding are generated using BERT with the embedding size of 1024.
(5) *Lexical and syntactic analysis*. The prepossessed data are translated into a sequence of tokens, and the syntactic context of each sentence is derived. To ensure that all inflected forms of the words are covered, this step conducts lemmatization and POS tagging.
(6) *Tokenization*. The purpose of tokenization is to chop the review sentences into multiple pieces referred to as "tokens" (e.g., "I like the old style" to "I," "like," "the," "old," "style").
(7) *Encoder*. In the BERT model, the inputs are pure text words. The encoder is used to transfer the input words into hidden high-dimensional vectors.
(8) *Decoder*. In addition to an encoder, the BERT model also requires a decoder to transfer the hidden high-dimensional vectors back into words.
(9) *Loss*. The model performance is measured as the cross entropy of the predicted labels versus the actual labels.

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(10) *Padding*. Most deep learning models, including BERT, require inputs of the same fixed size. BERT allows a maximum of 512 input sequence length. After identifying the longest input sequence, it is then necessary to fill the remainder of the shorter sequences up to that fixed length (usually with 0s).
(11) *Imbalanced classification*. Since over 90% of the generated labels are background words (i.e., "NA"), it is necessary to structure the softmax layer of the BERT model such that it avoids the *accuracy paradox*.
(12) *POS tag*. In NLP, POS tagging refers to the process of associating each word with a particular POS based on its definition and the context of the NLP task (e.g., "sole" is a noun in the context of shoes/sneakers).

3.2 The BERT Model

The BERT base model comprises an encoder with 12 transformer blocks, 12 self-attention heads, and a hidden layer of size 768. BERT takes a maximum input sequence with 512 tokens and outputs the labels of the words within that sequence. The sequence of the BERT input begins with [CLS] that includes special embeddings and another token [SEP] at the end of the sequence. Since the NLP task in this work is classification, a softmax layer is added atop BERT to predict the labels out of $C = \{$"NA," "S," "A," and "D"$\}$ as follows:

$$p(c_i|h) = \sigma(\Theta h)_j, \quad \forall j \in C \tag{1}$$

where σ is the softmax function, Θ is the task-specific parameters matrix, h is the hidden vector, and c_i is the predicted class. A BERT model with two additional convolutional neural network (CNN) layers is developed in this article (Figure 2), and its performance is compared with a baseline uncased BERT model.

The Fine-Tuning Strategy. To fine-tune BERT for the classification task, several factors must be taken into consideration: (1) *Preprocessing of long text*. Since our dataset involves user reviews, the maximum sequence length for BERT is set to 128. (2) *Layer selection*. The standard BERT model consists of an embedding layer, a 12-layer encoder, and a pooling layer. To select the most effective layers for the text classification task, a softmax layer is added atop BERT to carry out the classification task. (3) *Overfitting avoidance*. To avoid overfitting, an efficient optimizer with an appropriate learning rate is desired, which is identified by experimenting various learning rates. With these considerations, the BERT model is developed and implemented through the following steps:

- *Preprocessing*. BERT requires the input data to be in the special format discussed earlier. Upon tokenization, the dummy token [CLS] is added to the left side of each sentence. The sentences shorter than the maximum sequence length of 128 are then padded with zeros.
- *Selecting and deploying a pretrained BERT model*. The BERT pretrained model is open source and can be freely accessed online. There are several versions of BERT trained on different corpora. The uncased BERT model is used as baseline in our work.
- *Formulating the loss function for imbalanced classification*. Cross entropy has been widely used as loss function in various NLP tasks. The traditional cross entropy function calculates two probabilities that represent the actual class and the predicted class. The cross entropy function used in our work is as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} y_{o_i,c_i} \log\left(p_{o_i,c_i}\right) \tag{2}$$

where i is the ith sample in a minibatch, y is a binary indicator representing whether the class label c_i is the correct classification of observation o_i, p is the predicted observation o_i for class c_i, and N is the minibatch size. Since most reviews are written in informal language and by a diverse group of users, a major chunk of the input corpus involves background words (i.e., "NA"). The zero paddings used to equalize the length of the input sequences increase the number of background words even further. Specifically, around 90% of our input data are "NA." Training the BERT-NER model on these data without modification of the loss function would cause a phenomenon known as the *accuracy paradox*: the output would almost only include "NA" and the model precision would falsely go beyond 90%. To avoid this problem, a weighted loss function is applied as follows:

$$\mathcal{L}_w = -\frac{1}{N}\sum_{j=1}^{|C|} w_j \sum_{i}^{N} y_{o_{ij},c_{ij}} \log\left(p_{o_{ij},c_{ij}}\right) \tag{3}$$

where w_j is the weight of class label j, for all $j \in C$, C is the set of predefined classes {"NA," "S," "A," "D"}, and c_{ij} and o_{ij} are the predicted and observed labels for the ith word in class j, respectively. The weights w_j must be assigned commensurate with the appearance ratio of each class in the input corpus as well as with their importance to the NLP task. In our work, the weights have been assigned empirically based on the best observed performance (see Sec. 4.1 for details).
- *Choosing an optimizer*. The Adam optimizer [8] is chosen as the optimizer for BERT fine-tuning to avoid the limitation of stochastic gradient descent algorithms associated with maintaining a fixed learning rate for network weights updating. Adam calculates an exponential moving average of the recent magnitudes of the gradients for the network weights.
- *Designing the softmax layer*. The last layer of BERT is a vector with the dimension size of $[N, \mu, \lambda]$, where N is the batch size, μ is the maximum user-defined length, and λ is the size of the hidden layer. In our BERT model, the last layer is design with the following setting: [20, 128, 768]. For the sake of performance comparison, our strategy in this step is to experiment two approaches for connecting the softmax layer: (1) Put a linear layer atop BERT, which directly calculates the classification probabilities. (2) Put two additional convolutional layers atop BERT. The reason behind the second approach is to see whether performance can be improved with smaller labeled training data. The parameters setting for the convolutional layer is as follows: padding = 0, since no addition hidden layers are needed for the CNN, and stride = 5, kernel size = 3, to speed up the training process.
- *Hyperparameter setting*. The maximum length of the input sequence is 128, and the number of epochs is 200. The best model performance is kept during the training process. Over 20 different parameters settings are tested. Details are provided in Sec. 4.

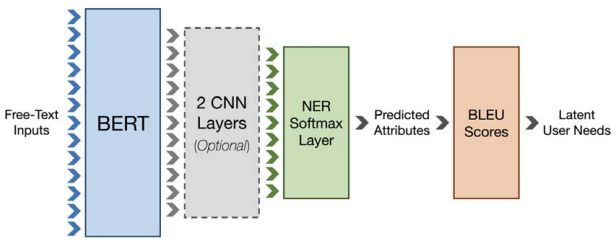


Fig. 2 The BERT-NER network architecture for elicitation of attribute-level user needs

3.3 The Named Entity Recognition Layer

This layers is the last softmax layer added on the top of the BERT and convolutional layers. Supervised NER is applied for performing

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the classification task. The NER layer is connected to the last layer of BERT, which has a dimension size of [20, 128, 768]. The output size of the NER layer is [20,128,4]. The last dimension of the NER output comprises the probabilities for the four classes of labels, where the one with the largest probability value is selected. The padding numbers added to the inputs do not influence the model performance because their outputs are all zeros. In our particular classification task, the paddings exacerbated the imbalance of our data because all paddings were labeled as "NA." This issue is addressed in Sec. 3.2. In the BERT-NER network with convolutional layers, it is required to first calculate the output dimension of the convolutional layer. The following parameters need to be set: the batch size N, the number of input channels (128 in our case), the number of output channels (4 in our case), the lengths of the input vector Γ_{in}, and the output vector Γ_{out}. The dimension changing rule of the convolutional layer is expressed as follows:

$$\Gamma_{\text{out}} = \frac{\Gamma_{\text{in}} + 2\pi - d(k-1) - 1}{s} + 1 \tag{4}$$

where π is the padding size of the convolutional layer, d is the dilation (the space between the kernel elements), k is the kernel size, and s is the stride size. The output shape of the last convolutional layer in our case is [20, 128, 256]. Dropout is not used between the convolutional layers because our goal is to learn more from the less data.

3.4 Performance Metrics

Four metrics are utilized to measure the performance of the BERT-NER methodology.

Average Weighted Loss. The losses averaged across observations for each minibatch. Eq. (3) is used to measure the average weighted loss of the imbalanced classification task.

Precision. The measure of how many of the selected words in a minibatch are correctly labeled.

$$\xi = \frac{1}{N}\sum_{i=1}^{N} w_j \sum_{j=1}^{|C|} \frac{|c_{ij} \cap o_{ij}|}{|c_{ij}|} \tag{5}$$

where N is the minibatch size, i is the sample in a minibatch, and c_{ij} and o_{ij} are the predicted and observed labels for the ith word in class j.

Recall. The measure of how many labels are correctly selected in a minibatch.

$$\tau = \frac{1}{N}\sum_{i=1}^{N} w_j \sum_{j=1}^{|C|} \frac{|c_{ij} \cap o_{ij}|}{|o_{ij}|} \tag{6}$$

F-score. The balance of precision and recall to provide a more realistic measure of the model performance. It is measured as the weighted harmonic mean of precision and recall.

$$\phi = \left(\frac{\xi^{-1} + \tau^{-1}}{2}\right)^{-1} \tag{7}$$

3.5 Eliciting User Needs With BLEU

The BLEU score is a benchmark to measure the fluency of a language model's output [52] usually in machine translation applications. The BLEU score usually ranges in [0, 1], where 1 represents 100% fluency. In our work, the BLEU score is utilized to elicit attribute-level user need expressions based on the results of the BERT-NER algorithm. To this end, all the words in the input text are POS tagged, and the words with tags "JJ" (adjective), "NN" (noun), and "VB" (verb) are selected. The selected words (adjective/noun/verb) include singular, plural, comparative, mass, past tense, gerund participle, nonthird person, singular present, and third person singular present. The BLEU scores of all the selected "JJ-NN" and "VB-NN" combinations are then calculated based on the user reviews (raw input data) data as a reference. The basic assumption is that the targeted word combinations are expressions of the user's needs, preferences, or opinions associated with different standard or nonstandard attributes of the product. It must be noted, however, that the authors do not suggest this assumption as a general rule and the developed methodology is not limited to the aforementioned combinations.

The BLEU score requires two sets of parameters: references and prediction. In our study, the training corpus is utilized as the reference, which includes the position of each word in the training corpus. The BLEU score counts the correct position of the output and then calculates a normalized score by assigning different weights to those positions. The BLEU score for n-gram candidates is calculated as follows:

$$\mathcal{B} = \mathcal{P}\exp\left[\sum_{m=1}^{M} \gamma_m \log R_m\right] \tag{8}$$

where $\mathcal{P}$ is the *brevity penalty* [52], M is the maximum length of candidates output, γ_m is a positive weight coefficient ($\sum_m \gamma_m = 1$), and R_M is the n-gram precision. The BLEU score is applied in our study to elicit attribute-level user needs as follows: (1) Select the 1000 most frequent attributes from the BERT-NER results. (2) Identify all the adjective–noun and verb–noun combinations as candidate user need expressions. (3) Calculate the BLEU score for each pair with reference to the entire training corpus. (4) Identify meaningful user needs with high BLEU scores.

4 Experiments and Results

This section tests and validates the performance of the BERT-NER methodology by experimenting on a large dataset scraped from an online retail store for apparel and footwear. The consumer products studied are sneakers.

4.1 Design of Experiments. Details of data collection, parameter setting, and benchmarks are provided in this section. The source codes and the attribute lexicon are available in online.[2]

- *Data collection and cleaning*. The sneaker dataset comprises all user reviews for sneakers posted on the online retail store before July 31, 2019. The total number of reviews is 23,564. The dataset provides information on multiple aspects including the customer location, gender, overall ratings, the title, and the body of the review. In our experiments, only the review titles and bodies were captured, and the remainder of the data was discarded. The raw data included lots of typos, grammar errors, and slang, which were directly imported into the algorithms without correction.
- *Labeling*. For fine-tuning the BERT model, a small labeled dataset is required. The labeling task was done by two individual NLP researchers in parallel. Four types of labels were assigned including "NA" (background word), "A" (attribute word), "S" (sentiment word), and "D" (attribute description word). The dataset is extremely imbalanced as discussed earlier (around 90% "NA"). The imbalance was addressed using the weighted cross entropy loss presented in Eq. (3) by assigning a relatively low weight to "NA" and also by labeling the padding characters as "NA." The weights were assigned as follows: $w_{\cdot\text{A}} = 0.45$, $w_{\cdot\text{NA}} = 0.05$, and $w_{\cdot\text{S}} = w_{\cdot\text{D}} = 0.25$. In the BERT-NER with CNN layers, with the best performance learning rate of 1e-5, we chose a different

[2] https://github.com/hanyidaxia/NER_BERT

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weights set through experiment: $w_{\cdot A \cdot} = 0.38$, $w_{\cdot NA \cdot} = 0.1$, and $w_{\cdot S \cdot} = w_{\cdot D \cdot} = 0.26$.

- *Programming*. The hyperparameters were set as follows: learning rate = 1e-4, 1e-5, 1e-6, and 2e-5; epochs = 100, 200; start up with l2 normalization; batch size = 20; dropout rate = 0; max sequence length 128.
- *Running the experiments*. For different combinations of parameter setting, the best performance for weighted loss, precision, recall, and F-score were saved during the training process. Of the 130 labeled data points, 90 were used for training, 20 for validation, and 20 for testing.
- *Benchmarks*. The performance of our BERT-NER model with CNN layers was compared against the baseline uncased BERT with a single layer, which has shown state-of-the-art performance in several NLP tasks. The motivation for adding the CNN layers stems from the fact that our training dataset is imbalanced and small.
- *Impact of imbalanced classification*. Almost all of our models showed a similar behavior: high loss with high precision and recall. This is due to the extreme imbalance of the data. Hence, the output of our original model (without weighted cross entropy loss) was imbalanced as well. Most of the words were labeled "NA," and the precision was unusually high. To avoid this behavior, the labels were assigned different weights and Eq. (3) was used for loss calculation.

4.2 Results. The performance results of the BERT-NER model with CNN layers and the benchmark uncased BERT-NER model (baseline) with different learning rate are presented in Table 1.

Model Performance. The weighted loss is the most important benchmark in our experiments due to the imbalanced nature of the data. The results presented in Table 1 indicate that the best performance is achieved with the learning rate of 1e-5 in both the BERT-NER baseline model and the BERT-NER model with CNN layers. Our observation during the training process was that as the learning rate decreases, the performance first increases and then drops. The potential reason for this behavior is that although decreasing the learning rate helps with updating the network weights, choosing a too small learning rate would cause the model to overfit. Another observation was that the two additional CNN layers significantly enhance the model performance for learning rates around 1e-6. The best performance for all parameter settings was achieved with a dropout rate of 0.5. The number of epochs used for training the models was set to a quite large value (200) to avoid overfitting.

Table 1 The BERT-NER model performance with different learning rates

Metric	Learning rate	BERT-NER baseline	BERT-NER-CNN
Weighted loss ($\mathcal{L}_w$)	1e-4	0.76	0.65
	1e-5	0.44	0.44
	2e-5	0.47	0.49
	1e-6	0.88	0.67
Precision (ξ)	1e-4	0.83	0.86
	1e-5	0.92	0.94
	2e-5	0.90	0.92
	1e-6	0.88	0.73
Recall (τ)	1e-4	0.88	0.91
	1e-5	0.89	0.92
	2e-5	0.91	0.96
	1e-6	0.83	0.74
F1 score (ϕ)	1e-4	0.86	0.88
	1e-5	0.90	0.93
	2e-5	0.90	0.93
	1e-6	0.86	0.73

Eliciting Attribute-Level User Needs. Having identified the best performance and learning rate of the BERT-NER model with CNN layers, 6160 unique attribute word candidates were extracted from the review corpus. As expected, the majority of those attributes words were not standard attributes of the product under study. Thus, the BLEU score analysis was applied for eliciting the attribute-level needs. To this end, the 6160 extracted attributes were sorted by their *cosine similarity* to our attribute lexicon, and the top 500 attributes that are not in the predefined attribute lexicon were selected for further analysis (this value could be modified in other cases). The cosine similarity is a benchmark to calculate similarity between words. After embedding all words into high-dimensional vectors using BERT, word similarity can be measured as the distance between those vectors. For each pair of word vectors v_1 and v_2, the cosine similarity was calculated as follows:

$$\mathcal{S} = \frac{\overrightarrow{v_1}\,\overrightarrow{v_2}}{\|\overrightarrow{v_1}\|\,\|\overrightarrow{v_2}\|} \tag{9}$$

The top 20 attributes identified in our experiments based on the aforementioned process include: "outsole," "color," "tear," "stripe," "structure," "alignment," "overlap," "tech," "carbon," "weight," "tube," "core," "reaction," "connectivity," "powering," "shipment," "velocity," "tightness," "looseness," and "fashionability." The quality of the extracted attributes depends highly on the model performance as well as the quality of the attribute lexicon. Thus, some of the attributes may not be good representatives of the attributes of a shoe (e.g., "tech" or "tube").

As presented in Sec. 3.5, the words with JJ-NN and VB-NN POS tag combinations were picked, and all combinations of those attributes were generated along with their respective BLEU scores. The BLEU score is an indicator of how realistic a JJ-NN or VB-NN pair is with reference to the actual reviews written by the users. The selected JJ-NN and VB-NN combinations require further interpretation to uncover the user needs and, of course, some extracted combinations may not contain any useful information. Illustrative examples are provided below to further elaborate on how the BERT-NER methodology works.

Example 1. JJ-NN combinations

> "The uncaged Ultra Boost offers a fit goes perfectly with the foot Also the sock upper really makes it very easy to slip them on and gives them a very nice look." JJ-NN: nice slip.
>
> "Honestly I love this shoe. I got it for Christmas. I love how easy it is to get it on and off. But when we first tried it on my heel would easily slide up and down in the back." JJ-NN: easy slide.
>
> "I really like the shoe. It fits very well easy to slip on and off just the way I need it to." JJ-NN: easy slip.
>
> "I love these shoes. The stretchy fabric and the neck on the shoes are amazing and very easy to slip into." JJ-NN: easy neck. ■

The aforementioned examples demonstrate the needs of users for sneakers that are easy to put on and off. Although some brands already offer categories of shoes called "slip-ons," this could still trigger new, transformative design ideas to make all types of shoes and sneakers easy to put on/off in the future.

Example 2. JJ-NN combinations

> "I am a female 47 years old runner. These shoes are the best running shoes I've had in a long time. They are perfect for my flat feet I usually struggle finding the perfect shoe but these were absolutely amazing I can't wait to buy them in every color." JJ-NN: perfect flat.
>
> "I have had a fresh pair of Stan Smiths as part of my wardrobe for the past 15 years They look great with jeans chinos and shorts they are very versatile. They are also suitable for leg day squats and dead lifts due to the flatness of the sole." JJ-NN: suitable flatness. ■

These users suffer from flat feet (also known as fallen arches)—a common condition spanning 20–30% of the general population. Note that the word "flat" has been extracted as an attribute referring

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to the flatness of the insole. This is an example of user needs for personalizable, adjustable, or potentially smart insoles that could respond or adapt to the flatness of their feet. The following example is in a similar vein, expressing the need for personalizable/adjustable/potentially smart insoles:

Example 3. VB-NN pair

> "I really liked the design but once I received the shoe it fitted a little snug Had to remove the insole for it to fit Also the tab at the back rubbed on the back of my feet." VB-NN: remove insole. ■

The results of the proposed BERT-NER methodology may require further manual postprocessing to draw useful insights from the extracted JJ-NN or VB-NN combinations, which may in turn impose considerable amounts of cognitive load on the designer. The methodology may also yield several meaningless/useless combinations such as "peeled logo," "incompatible pain," or "easy shoe," which require manual clean-up after running the experiments. These are among the foremost limitations of this work to be addressed in future research, as discussed in Sec. 5.

5 Conclusions and Future Research Directions

This article proposed a novel methodology for automated and large-scale elicitation of attribute-level user needs from online reviews. Building on advanced NLP research on pretrained language models, the methodology is anticipated to save significant amounts of time and effort for data preparation and reduce the need for hand-engineered expert systems for need finding from user reviews. The developed methodology also enables efficient, continual updating of the need finding results as more users publish their feedback and opinions on e-commerce platforms on a daily basis. The advantages of the BERT-NER methodology for large-scale elicitation of attribute-level user needs include the following:

- *Efficiency and scalability*. The use of pretrained language models such as BERT reduces the need for large manually labeled data. Specifically, the experiments conducted in this work only utilized 130 labeled data, which is orders of magnitude smaller than traditional deep learning datasets that usually comprise thousands of labeled data. All components of the BERT-NER methodology are packaged in a structured fashions and can be quickly modified and applied to any need finding problem. The methodology also reduces the need for conducting extensive market studies, focus groups, interviews, or lead user analyses. The only input required is free-text, user reviews. Accordingly, the required investments in of human power, budget, and space for conducting large-scale need finding studies can be dramatically reduced.
- *Automated and large-scale need finding*. The BERT-NER methodology extracts an exhaustive list of candidate attribute words. The list is then further refined in a top-down fashion using the BLEU score to elicit expressions (i.e., JJ-NN and VB-NN combinations) that represent the expressed needs of users in the review corpus. This is a significant step toward enabling automated and large-scale need finding, which, compared to lead user-based approaches, can potentially extract more informative and potentially transformative insights to inform the design process.
- *Modular structure*. The BERT-NER network is composed of independent modules for BERT, CNN, NER, and BLEU score analysis (see Fig. 2). Such modular structure enables flexibility for independent/parallel improvement of different modules with new add-on features for handling different NLP tasks.

To fully realize the potentials of the proposed methodology, however, future research will be required across three frontiers, including methodology, validation, and generalization to more complex, nonobvious, and hard-to-identify user needs, or *latent needs*.

Methodology. Further research will be needed to optimize the BERT-NER network architecture for reducing loss and improving precision. A comprehensive study of all potential POS-tags representing latent need expressions will also be necessary to improve recall. Major technical limitations of the BERT-NER methodology to be improved in future work are as follows:

- *High loss*. The input free-text reviews are noisy and imbalanced. As explained earlier, this extreme imbalance leads to high precision with high loss—a phenomenon known as the accuracy paradox. The authors speculate that this deficiency could be overcome by increasing the size of labeled training data.
- *Low recall for latent needs*. The underlying assumption of the BLEU analysis is that user needs are typically expressed in the form of JJ-NN or VB-NN combinations. In practice, however, user needs may be expressed in other forms of POS-tag combinations, and thus other potentially relevant POS tag combinations should be tested for user need elicitation.
- *Potential information loss due to long network structure*. The BERT-NER network (Fig. 2) comprises several submodels for BERT (with or without CNN layers), NER, POS tagging, and the BLEU score calculation. Every submodel has a loss, which in turn may cause information loss and make the output very noisy. This problem can be addressed by improving the architecture of each submodel.

Validation. It is unclear whether and how the proposed methodology will impact the performance of the design team in finding meaningful and informative user needs in practice. Future research must conduct extensive human subject studies in controlled laboratory environments to measure the hypothesized difference between the performances of a design team using the BERT-NER results and another design team reading reviews directly from e-commerce platforms. Moreover, the JJ-NN and VB-NN combinations yield brief expressions of user needs that may require further interpretation by designers to uncover the underlying user needs. A considerable portion of the results also contains expressions that are either meaningless (e.g., "easy cool") or not informative (e.g., "extra wool"). Future research must devise new mechanisms to measure how informative each identified user need is for the designer. That is, even with a highly accurate methodology capable of accurately extracting all user needs, it is unclear whether the designer will be "surprised" by the results or how much she will learn from them.

Latent Need Finding. The proposed methodology can potentially contribute to the elicitation of *latent* user needs that "may be nonobvious and very difficult to identify, and sometimes customers may not be consciously aware of them but are unexpectedly delighted if they are satisfied or disgusted if they are not satisfied" [39]. This can be done by shifting the need finding strategy from asking users for their desired "solutions" to understanding what "job" they would like to accomplish with the product [85]. Previous research on the identification of implicit and nonobvious user needs can be classified into two categories [57]: lead user based and context based. The former typically applies case-based or analogical reasoning methods in conjunction with the sentiment analysis to infer user needs based on explicit needs and extraordinary use cases [86]. It also involves context-based design approaches to explicitly model and measure how usage context influences user choices and product performance [87]. The latter, on the other hand, involves studying lead users [88] (e.g., disabled users) to see how their opinions would influence the product design.

Building on these and other related previous studies, the BERT-NER methodology can potentially enable an automated, quantitative, and viable mechanism for large-scale elicitation of latent user needs. Future research should investigate the mechanisms and POS combinations that users apply to express the job they would like to accomplish rather than the POS combinations

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used to express their opinions about certain attributes of the product (e.g., "personalizable, adjustable, or potentially smart insoles"). The identification of such mechanisms and POS combinations would be essential to further the development of the BERT-NER methodology for large-scale latent need finding to potentially provide innovative insights to the design team.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request. The authors attest that all data for this study are included in the paper.

Nomenclature

d = dilation space between the kernel elements
h = hidden vector
i = ith sample in a minibatch
j = class label ($j \in C$)
k = Kernel size
r = effective reference corpus length
s = stride size
w = weight vector
C = set of predefined classes {"NA," "S," "A," "D"}
M = maximum length of candidates output
N = minibatch size
$\mathcal{B}$ = BLEU score for n-gram candidates
$\mathcal{L}$ = cross entropy loss
$\mathcal{P}$ = brevity penalty
$\mathcal{S}$ = cosine similarity of two vectors
$\mathcal{L}_w$ = weighted cross entropy loss
$\overrightarrow{v_1}$ = vector
$\overrightarrow{v_2}$ = vector
c_i = predicted label for observation o_i
c_{ij} = predicted label for observation o_{ij}
o_i = observation for predicted class label c_i
o_{ij} = observation for predicted class label c_{ij}
γ_m = positive weight coefficient
Γ_{in} = length of input vector
Γ_{out} = length of output vector
Θ = task specific matrix
λ = size of hidden layer
μ = maximum user defined length of a deep learning network
ξ = precision of the model
π = padding size of the convolutional layer
σ = softmax function
τ = recall
ϕ = F1 score

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